

In []: *# Enter your name: Elijah Oliver*

In [2]: *# Run this cell as it is to import the needed libraries.*
Note: For the imports work correctly, you must
first install the libraries as discussed here: <https://youtu.be/nvx>

```
import geopandas as gpd
import pandas as pd
import matplotlib.pyplot as plt
import contextily
import mapclassify
import folium
import fsspec
```

In [3]: *# Run this cell as it is to load the GeoDataFrame.*
Note: This data set contains the geometry shapes for US counties.

```
county_data = "https://github.com/babdelfa/gis/blob/main/counties_geom"

with fsspec.open(county_data) as file:
    county_shapes = gpd.read_file(file)

# Subset the data columns of interest from county_shapes GeoDataFrame
county_shapes = county_shapes[['FIPS_BEA', 'geometry']].copy()

# View first five records
county_shapes.head()
```

Out [3]:

	FIPS_BEA	geometry
0	1001.0	POLYGON ((-86.41312 32.70739, -86.41304 32.687...
1	1003.0	POLYGON ((-87.76515 31.29735, -87.76360 31.297...
2	1005.0	POLYGON ((-85.35541 32.14717, -85.32444 32.147...
3	1007.0	POLYGON ((-87.06574 33.24691, -87.04483 33.246...
4	1009.0	POLYGON ((-86.48693 34.26021, -86.45288 34.259...

```
In [4]: # Run this cell as it is to load the DataFrame containing COVID-19 data
# Note: This data set represents reported cumulative COVID-19 cases for
# each US county (i.e., Admin2) as of March 9, 2023

covid_data = "https://github.com/CSSEGISandData/COVID-19/raw/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_cases.csv"
df = pd.read_csv(covid_data)

# Subset the data columns of interest from df
df = df[['UID', 'iso2', 'FIPS', 'Admin2', 'Province_State', 'Lat', 'Long_']

# Rename column
df.rename(columns={"3/9/23" : "cases"}, inplace=True)

# View first five records
df.head()
```

Out [4]:

	UID	iso2	FIPS	Admin2	Province_State	Lat	Long_	cases
0	84001001	US	1001.0	Autauga	Alabama	32.539527	-86.644082	19790
1	84001003	US	1003.0	Baldwin	Alabama	30.727750	-87.722071	69860
2	84001005	US	1005.0	Barbour	Alabama	31.868263	-85.387129	7485
3	84001007	US	1007.0	Bibb	Alabama	32.996421	-87.125115	8091
4	84001009	US	1009.0	Blount	Alabama	33.982109	-86.567906	18704

In [5]: *Join the GeoDataFrame and DataFrame together into gdf using the common data. The column labels differ.*

```
gdf = county_shapes.merge(df, left_on='FIPS_BEA', right_on='FIPS') # ---
gdf.head()
```

Out [5]:

	FIPS_BEA	geometry	UID	iso2	FIPS	Admin2	Province_State	Lat	Long
0	1001.0	POLYGON ((-86.41312 32.70739, -86.41304 32.687...	84001001	US	1001.0	Autauga	Alabama	32.539527	-86.64401
1	1003.0	POLYGON ((-87.76515 31.29735, -87.76360 31.297...	84001003	US	1003.0	Baldwin	Alabama	30.727750	-87.72201
2	1005.0	POLYGON ((-85.35541 32.14717, -85.32444 32.147...	84001005	US	1005.0	Barbour	Alabama	31.868263	-85.38711
3	1007.0	POLYGON ((-87.06574 33.24691, -87.04483 33.246...	84001007	US	1007.0	Bibb	Alabama	32.996421	-87.12511
4	1009.0	POLYGON ((-86.48693 34.26021, -86.45288 34.259...	84001009	US	1009.0	Blount	Alabama	33.982109	-86.56791

In [6]: *# Subset gdf to include only Texas counties data.*

```
tx = gdf[gdf['Province_State']=='Texas'].copy() # --- complete the syn
```

```
In [7]: # Permanently rename columns in tx as:
#       Admin2 to County
#       cases to Total Cases

tx.rename(columns={'Admin2': 'County', 'cases': 'Total Cases'}, inplace=True)
tx.head()
```

Out [7]:

	FIPS_BEA	geometry	UID	iso2	FIPS	County	Province_State	Latitude
2487	48001.0	POLYGON ((-95.42851 32.08447, -95.42411 32.083...	84048001	US	48001.0	Anderson	Texas	31.815347
2488	48003.0	POLYGON ((-102.99628 32.52315, -102.61377 32.5...	84048003	US	48003.0	Andrews	Texas	32.304686
2489	48005.0	POLYGON ((-94.86580 31.52696, -94.86406 31.525...	84048005	US	48005.0	Angelina	Texas	31.254576
2490	48007.0	MULTIPOLYGON (((-96.85716 28.11556, -96.84538 ...	84048007	US	48007.0	Aransas	Texas	28.105562
2491	48009.0	POLYGON ((-98.43096 33.83596, -98.42358 33.836...	84048009	US	48009.0	Archer	Texas	33.615706

```
In [8]: # Subset tx to only include the three data columns:
#       County, Total Cases, and geometry

tx = tx[['County', 'Total Cases', 'geometry']].copy() # --- complete t
tx.head()
```

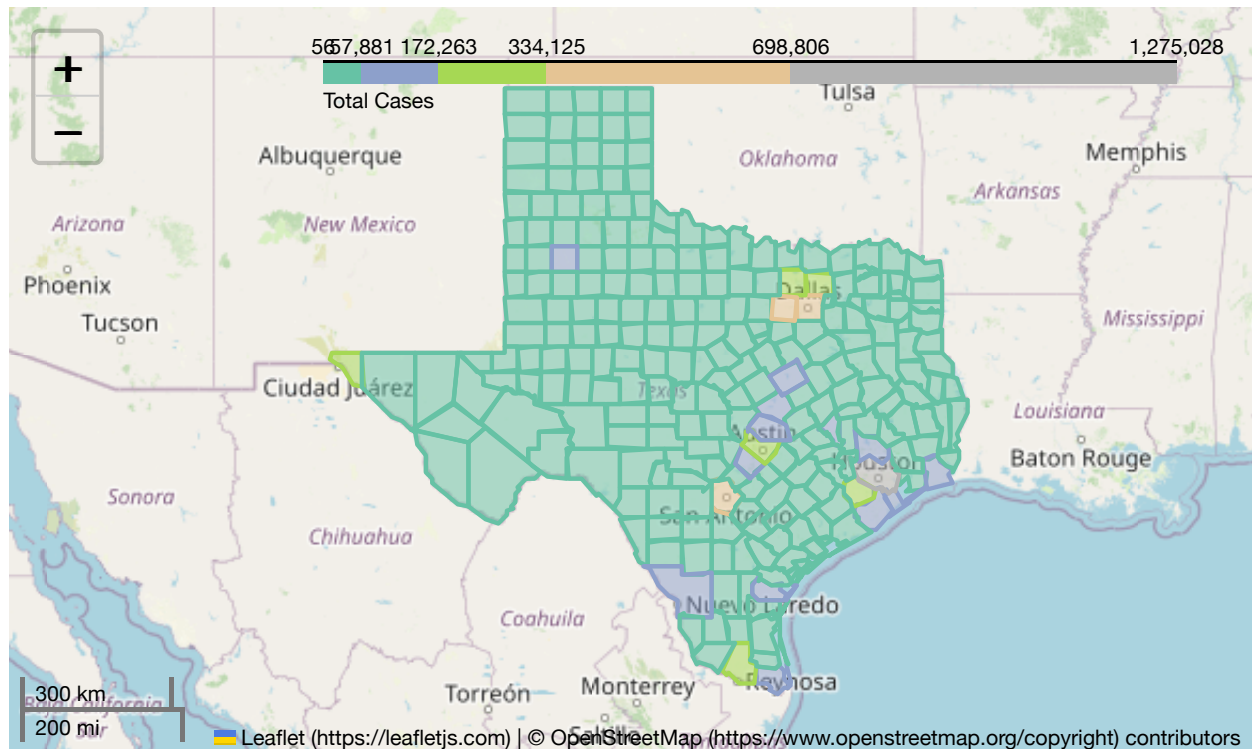
Out[8]:

	County	Total Cases	geometry
2487	Anderson	10975	POLYGON ((-95.42851 32.08447, -95.42411 32.083...
2488	Andrews	5093	POLYGON ((-102.99628 32.52315, -102.61377 32.5...
2489	Angelina	17089	POLYGON ((-94.86580 31.52696, -94.86406 31.525...
2490	Aransas	5550	MULTIPOLYGON (((-96.85716 28.11556, -96.84538 ...
2491	Archer	2646	POLYGON ((-98.43096 33.83596, -98.42358 33.836...

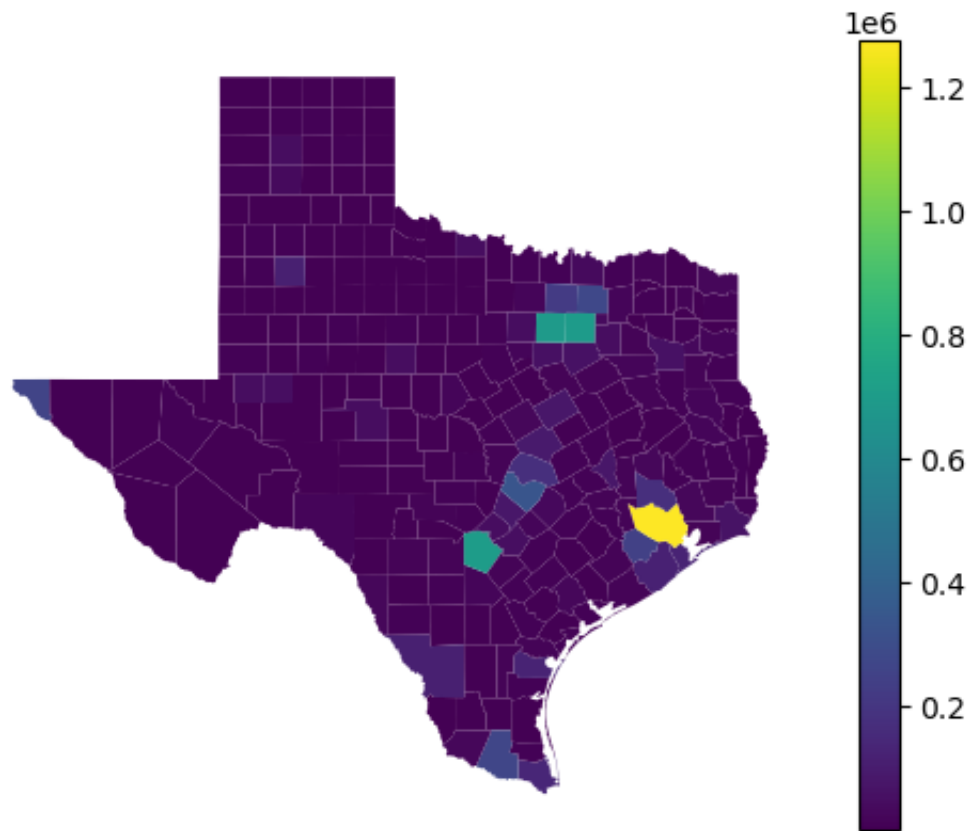
```
In [9]: # Display an interactive choropleth map of Total Cases for Texas (tx)

tx.explore(column='Total Cases', cmap="Set2", legend=True, scheme='Nat
```

Out[9]:



```
In [10]: # Run this cell as it is to display a non-interactive choropleth map of  
tx.plot(column="Total Cases", legend=True) # plots the geodataframe to  
plt.axis("off")  
plt.show()
```



In []: